

Tianyi Chen

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Education

Johns Hopkins University <i>Ph.D. at Applied Math and Statistics Department</i>	Baltimore, Maryland Sep 2021- May 2026
Johns Hopkins University <i>Master of Science in Engineering</i>	Baltimore, Maryland Aug 2019 - May 2021
Renmin University of China <i>Bachelor of Science</i>	Beijing, China Sep 2015 - June 2019

Conference Papers

Helm, H., **Chen, T.**, McGuinness, H., Lee, P., Duderstadt, B., & Priebe, C. E. (2025). *Toward a digital twin of U.S. Congress*. arXiv: 2505.00006. Findings at Association for Computational Linguistics Conference.

Journal Papers

Chen, T., Kiasari, M. S., Yu, S., Park, Y., Athreya, A., Lyzinski, V., Priebe, C. E., & Lubberts, Z. (2026). *Vertex misalignment and changepoint localization in network time series*. arXiv: 2604.20072. Submitted to Journal of the American Statistical Association.

Chen, T., Lubberts, Z., Athreya, A., Park, Y., & Priebe, C. E. (2024). *Euclidean mirrors and first-order changepoints in network time series*. arXiv: 2405.11111. Revised and resubmitted to Electronic Journal of Statistics.

Chen, T., Park, Y., Saad-Eldin, A. et al (2023). *Discovering a change point and piecewise linear structure in a time series of organoid networks via the iso-mirror*. Applied Network Science.

Priebe, C. E., Shen, C., Huang, N., & **Chen, T.** (2021). *A simple spectral failure mode for graph convolutional networks*. IEEE Transactions on Pattern Analysis and Machine Intelligence.

Wang, F., **Chen, T.**, Chang, Q., Kao, Y. W., Li, J., Chen, M., ... & Shia, B. C. (2021). *Respiratory diseases are positively associated with PM2.5 concentrations in different areas of Taiwan*. PLOS One, 16(4), e0249694.

Chen, T., Chen, Y., Gao, J., Gao, P., Moon, J. H., Ren, J., ... & Woolf, T. B. (2021). *Machine learning to summarize and provide context for sleep and eating schedules*. bioRxiv, 2020-12.

Awards and Fellowships

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| – Nominated as Siebel scholar by Applied Mathematics and Statistics Department | June 2025 |
| – IMSI "Statistics Meets Tensors" workshop travel fund | May 2025 |
| – Honorable Mention Audience Choice Award at the DSAI End of Semester Social for presentation of "Toward a digital twin of US Congress." | |
| | May 2025 |
| – Teaching Fellow Recognition Award | May 2025 |
| – Duncan Fund for travel | |

- Mathematical Institute of Data Science Fellowship Spring 2022

Research In Progress

Human Brain Organoid Analysis

May 2023-Present

Collaboration with *JHU Center for Alternatives to Animal Testing* and *Applied Physics Laboratory*

- Co-developed a pipeline that processes time series of brain organoid Micro Electrode Array (MEA) recording data into Euclidean representations.
- Explored the connectome’s dynamic evolution over time.

Swarm Movement Dynamics Analysis

Sep 2025-Present

Collaboration with *Naval Research Laboratory* and *Georgia Institute of Technology*

- Processed swarm movement video into Euclidean representations for changepoints analysis.
- Investigated the robustness of changepoint analysis to missing swarm alignment.

Teaching Experience

Teaching Assistant

- Course “Statistical Pattern Recognition Theory & Methods” Spring 2025
- Course “Statistics Theory” Fall 2021, Fall 2024
- Course “Time Series Analysis” Spring 2024
- Course “Applied Statistics and Data Analysis” Fall 2020, Fall 2022, Fall 2023
- Course “Applied Statistics and Data Analysis 2” Spring 2021
- Course “Intro to Optimization” Spring 2020

Instructor

- Master’s Program Statistics Review (7.5 hours) Summer 2023, Summer 2024

Awarded Teaching Fellow May 2025

Talks

- “Euclidean mirror and subsequent inference.” Nov 3 2025
JHU Junior Mathematical Institute of Data Science Fellow Seminar
- “Vertex alignment and first-order changepoint in time series of graphs.” Aug 13 2025
Science of Autonomy Annual Program Review, Virtual session
- “Vertex alignment and first-order changepoint in time series of graphs.” Aug 4 2025
Joint Statistical Meeting, Nashville
- “Toward a digital twin of US Congress.” April 30 2025
Symposium on Data Science and Statistics, Salt Lake City
- “Vertex alignment and first-order changepoint in time series of graphs” March 25 2025
JHU Applied Math and Statistics department Student Seminar
- “Euclidean mirror and first-order changepoint in network time series” Oct 23 2024
SIAM Conference on Mathematics of Data Science, Poster session, Atlanta
- “Euclidean mirror and first-order changepoint in network time series” May 20 2024
38th New England Statistics Symposium, invited talk, University of Connecticut
- “Organoid intelligence and time series of networks.” Sep 26 2023
JHU Applied Math and Statistics department Student Seminar

“A simple example for understanding swarms.”

Oct 11 2022

JHU Applied Math and Statistics department Student Seminar

“Understanding swarm: manifold learning on spectral representation of time series of graphs.”

JHU Applied Math and Statistics department Student Seminar

Oct 19 2021

Services

Member, Diversity Committee, Department of Applied Math and Statistics Sep 2024 – May 2025

Organizer, Special Seminar: “Converging Perspectives,” Department of Applied Math and Statistics
April 4, 2025

Volunteer, 7th and 9th International Forum on Statistics, Beijing

May 2016, Jul 2018